

Solving systems of linear D.E by elimination

2 or more ordinary DE that contains 2 or more dependent variable (unknown function) as (x or y) with respect to a single independent variable as t

ex:

$$\begin{cases} \frac{dx}{dt} = y \\ \frac{dy}{dt} = 3x \end{cases}$$

- get everything onto one side or to the left hand side and keep the constants if there are any on right hand side

$$\begin{cases} \frac{dx}{dt} - y = 0 \\ \frac{dy}{dt} - 3x = 0 \end{cases}$$

- set the $\frac{dx}{dt} \rightarrow Dx$ and $\frac{dy}{dt} \rightarrow Dy$ the D stands for $\frac{d}{dt}$

$$\begin{cases} Dx - y = 0 \\ Dy - 3x = 0 \end{cases}$$

- you can eliminate either x or y then add the two equations your goal here is to get it into one equation with one term

$$\begin{cases} D^2x - Dy = 0 \\ Dy - 3x = 0 \end{cases}$$

- we multiplied the first equation by D to get rid of Dy from both equations

$$D^2x - 3x = 0$$

$$D^2x - 3x = 0 \rightarrow x'' - 3x = 0$$

- we said D was just the derivative

$$m^2 - 3 = 0$$

$$m = \pm \sqrt{3}$$

- this is a case #1 so we will write the equation for it

$$x(t) = C_1 e^{-\sqrt{3}t} + C_2 e^{\sqrt{3}t}$$

• We got $x(t)$ now we need to find $y(t)$

but if we look at the given equations we see that: $\frac{dx}{dt} = y$ y is the derivative of $x(t)$

$$x(t) = C_1 e^{-\sqrt{3}t} + C_2 e^{\sqrt{3}t}$$

$$x'(t) = -\sqrt{3} C_1 e^{-\sqrt{3}t} + \sqrt{3} C_2 e^{\sqrt{3}t}$$

Plug this into the equation you should have y

$$y = -\sqrt{3} C_1 e^{-\sqrt{3}t} + \sqrt{3} C_2 e^{\sqrt{3}t}$$

#1)

$$\begin{cases} \frac{dx}{dt} = 2x - y \\ \frac{dy}{dt} = x \end{cases}$$

→

$$\begin{cases} Dx - 2x + y = 0 \\ Dy - x = 0 \end{cases}$$

• multiply equation #1 by $-D$

• this is a case #2

$$\begin{cases} -D^2x + 2x - Dy = 0 \\ Dy - x = 0 \end{cases}$$

$$-x'' + x = 0$$

→

$$-m^2 + 1 = 0$$

→

$$m = \pm 1$$

$$x(t) = C_1 e^t + C_2 t e^t$$

We found $x(t)$ we can find $y(t)$ by looking at the given eqn's so y would be equal to:

$$y = 2x - \frac{dx}{dt}$$

$$x(t) = C_1 e^t + C_2 t e^t$$

$$\frac{C_2 t e^t}{f} \frac{e^t}{g}$$

$$x(t) = C_1 e^t + C_2 e^t + C_2 t e^t$$

$$f \cdot g + f \cdot g'$$

$$C_2 e^t + C_2 t e^t$$

now plug into the equation to get $y(t)$:

$$y(t) = 2(C_1 e^t + C_2 t e^t) - (C_1 e^t + C_2 e^t + C_2 t e^t)$$

$$y(t) = \underline{2C_1 e^t} + \underline{2C_2 t e^t} - \underline{C_1 e^t} - C_2 e^t - \underline{C_2 t e^t}$$

$$y(t) = C_1 e^t + C_2 t e^t - C_2 e^t$$

#2)

$$\begin{cases} \frac{dx}{dt} = -y + t \\ \frac{dy}{dt} = x - t \end{cases}$$

→

$$\begin{cases} Dx + y = t \\ Dy - x = -t \end{cases}$$

• multiply the first equation by $-D$

$$\begin{cases} -D^2 x - \cancel{Dy} = -Dt \\ \underline{Dy - x = -t} \\ -D^2 x - x = -Dt - t \end{cases}$$

$$-x'' - x = -1 - t \rightarrow -x'' - x = 0 \rightarrow -m^2 - 1 = 0$$

$$m = \pm i$$

↑ this is a case #3

↑ you can take the derivative of this

$$x_c(t) = C_1 \cos(t) + C_2 \sin(t)$$

• this is a non-homogeneous equation so you need to use trial particular soln to find y_p and then plug into eqn

$$x_p = At + B$$

$$x'_p = A$$

$$x''_p = 0$$

$$-x'' - x = -1 - t$$

$$-\cancel{C_1} - (At + B) = -1 - t$$

$$-At - B = -1 - t$$

$$-At = -1, \quad -B = -1$$

$$A = 1, \quad B = 1$$

$$x_p = t + 1$$

Now use this equation to $x = x_c + x_p$ get

$$x = C_1 \cos(ct) + C_2 \sin(ct) + t + 1$$

You can use this equation to find y :

$$y = t - \frac{dx}{dt}$$

$$x = C_1 \cos(ct) + C_2 \sin(ct) + t + 1$$

$$x' = -C_1 \sin(ct) + C_2 \cos(ct) + 1$$

$$y = t - (-C_1 \sin(ct) + C_2 \cos(ct) + 1)$$

$$y(t) = t + C_1 \sin(ct) - C_2 \cos(ct) - 1$$

$$\begin{cases} x' = 3x - 4y + 6 \\ y' = 4x - 7y + 14t \end{cases}$$

→

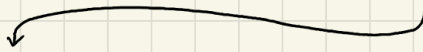
$$\begin{cases} (D-3)x + 4y = 6 \\ -4x + (D+7)y = 14t \end{cases}$$

multiply first equation by 4 and multiply the second equation by (D-3)

$$\begin{cases} 4(D-3)x + 4y = 24 \\ -4x + (D+7)y = 14t \end{cases}$$

→

$$\begin{cases} \cancel{4Dx} - \cancel{12x} + 4y = 24 \\ \cancel{-4Dx} + \cancel{12x} + (D-3)(D+7)y = 14t(D-3) \end{cases}$$



$$16y + (D-3)(D+7)y = 24 + 14t(D-3)$$

$$16y + (D^2 + 4D - 21)y = 24 + 14 - 42t$$

$$16y + y'' + 4y' - 21y = 38 - 42t$$

$$y'' + 4y' - 5y = 38 - 42t$$

$$m^2 + 4m - 5 = 0 \rightarrow$$

$$(m+5)(m-1)$$

$$m = -5, m = 1$$

• case #1

$$y_c = C_1 e^{-5t} + C_2 e^t$$

$$y_p = At + B$$

$$y_p' = A$$

$$y_p'' = 0$$

$$\cancel{0} + 4(A) - 5(At + B) = 38 - 42t$$

$$4A - 5At - 5B = 38 - 42t$$

$$\cancel{-5At} = \cancel{-42t}, \quad 4A - 5B = 38$$

$$A = \frac{42}{5}, \quad 4\left(\frac{42}{5}\right) - 5B = 38$$

$$B = -\frac{22}{25}$$

$$y_p = \frac{42}{5}t - \frac{22}{25}$$

→

$$y = y_c + y_p$$

→

$$y(t) = C_1 e^{-5t} + C_2 e^t + \frac{42}{5}t - \frac{22}{25}$$

$$y' = 4x - 7y + 14t$$

$$4x = y' + 7y - 14t$$

$$4x = (-5C_1 e^{-5t} + C_2 e^t + \frac{42}{5}) + 7(C_1 e^{-5t} + C_2 e^t + \frac{42}{5}t - \frac{22}{25}) - 14t$$

$$4x = \cancel{-5C_1 e^{-5t}} + \cancel{C_2 e^t} + \frac{42}{5} + 7\cancel{C_1 e^{-5t}} + 7\cancel{C_2 e^t} + \frac{294}{5}t - \frac{154}{25} - 14t$$

$$4x = 2C_1 e^{-5t} + 8C_2 e^t + \frac{56}{25} + \frac{224}{5}t$$

$$x(t) = \frac{1}{2} C_1 e^{-5t} + 2 C_2 e^t + \frac{14}{25} + \frac{56}{5} t$$

$$\begin{cases} \frac{dx}{dt} + \frac{dy}{dt} = e^t \\ -\frac{d^2x}{dt^2} + \frac{dx}{dy} = -x-y \end{cases} \rightarrow \begin{cases} Dx + Dy = e^t \\ -D^2x + Dx + x + y = 0 \end{cases}$$

• we will multiply the second equation by -D

$$-D \cdot \begin{cases} Dx + Dy = e^t \\ -D^2x + Dx + x + y = 0 \end{cases} \rightarrow \begin{cases} \cancel{Dx} + \cancel{Dy} = e^t \\ D^2x - D^2x - \cancel{Dx} - \cancel{Dy} = 0 \end{cases}$$

$$D^2x - D^2x = e^x$$

$$D^2x - D^2x = e^t \rightarrow x''' - x'' = e^t \rightarrow m^3 - m^2 = 0 \rightarrow m = \pm 0, m = 1$$

$$m^2(m-1) = 0$$

• this is a mix of case #2 and case #1

$$x_c = C_1 + C_2 t + C_3 e^t$$

A and C_1 ? No, repetition

A and C_2 ? No, repetition

A and C_3 ? yes, repetition

$$x_p = A t e^t$$

$$\frac{A t e^t}{f'g}$$

$$f'g + f \cdot g'$$

$$x_p' = (A)(e^t) \cdot (At)(e^t)$$

$$x_p' = (A)(e^t) \cdot \frac{(At)}{f} \frac{(e^t)}{g}$$

$$f'g + f \cdot g'$$

$$(A)(e^t) + (At)(e^t)$$

$$x_p'' = A e^t + A e^t + A t e^t$$

$$x_p'' = 2 A e^t + A t e^t$$

$$x_p'' = 2 A e^t + \frac{A t e^t}{f'g}$$

$$f'g + f \cdot g'$$

$$(A)(e^t) + (At)(e^t)$$

$$x_p''' = 2 A e^t + A e^t + A t e^t$$

$$x_p''' = 3 A e^t + A t e^t$$

$$x''' - x'' = e^t$$

$$(3Ae^t + Ate^t) - (2Ae^t + Ate^t) = e^t$$

$$3Ae^t + \cancel{Ate^t} - 2Ae^t - \cancel{Ate^t} = e^t$$

$$\cancel{Ae^t} = \cancel{e^t}$$

$$A = 1$$

$$x_p = Ate^t \rightarrow x_p = te^t$$

Plug everything into $x = x_c + x_p$:

$$x(t) = c_1 + c_2 t + c_3 e^t + te^t$$

$$\frac{dx}{dt} + \frac{dy}{dt} = e^t \rightarrow \frac{dy}{dt} = e^t - \frac{dx}{dt}$$

*you can plug everything in here and take the integral so you can get rid of $\frac{dy}{dt}$

$$x(t) = c_1 + c_2 t + c_3 e^t + te^t$$

$$x'(t) = c_2 + c_3 e^t + e^t + \underline{te^t}$$

$$\frac{dy}{dt} = e^t - (c_2 + c_3 e^t + e^t + te^t)$$

$$\frac{te^t}{f'g}$$

$$f'g + f \cdot g'$$

$$e^t + te^t$$

$$\int \frac{dy}{dt} = \int e^t - c_2 - c_3 e^t - e^t - \underline{te^t}$$

$$y(t) = e^t + tc_2 - c_3 e^t - e^t - te^t + e^t + c_4$$

$$u \cdot v - \int v \cdot du$$

$$u = t$$

$$dv = e^t$$

$$du = 1 dt$$

$$v = e^t$$

$$-te^t + \int e^t dt$$

$$-te^t + e^t$$

$$\begin{cases} \frac{d^2 x}{dt^2} = 4y + e^t \\ \frac{d^2 y}{dt^2} = 4x - e^t \end{cases} \rightarrow \begin{cases} D^2 x - 4y = e^t \\ D^2 y - 4x = -e^t \end{cases}$$

• multiply the first equation by D^2 and the second equation by 4

$$\begin{array}{l} D^2. \\ 4. \end{array} \begin{cases} D^2 x - 4y = e^t \\ D^2 y - 4x = -e^t \end{cases} \rightarrow \begin{cases} D^4 x - \cancel{4D^2 y} = D^2 e^t \\ \cancel{4D^2 y} - 16x = -4e^t \end{cases}$$

$$D^4 x - 16x = D^2 e^t - 4e^t$$

$$x'''' - 16x = e^t - 4e^t \rightarrow x'''' - 16x = -3e^t \rightarrow \begin{array}{cc} (m^2 - 4) & (m^2 + 4) \\ m = \pm 2 & m = \pm 2i \end{array}$$

• this consists of a case #1 and case #3

$$x_c = C_1 e^{-2t} + C_2 e^{2t} + C_3 \cos(2t) + C_4 \sin(2t)$$

$$x_p = A e^t$$

A and C_1 and C_2 ? No repetition

$$x_p'''' = A e^t$$

A and C_3 and C_4 ? No repetition

$$x'''' - 16x = -3e^t$$

$$(A e^t) - 16(A e^t) = -3e^t$$

$$-15 A e^t = -3e^t$$

$$A = \frac{1}{5}$$

$$x_p = \frac{1}{5} e^t$$

• use this equation to get x : $x = x_c + x_p$

$$x(t) = C_1 e^{-2t} + C_2 e^{2t} + C_3 \cos(2t) + C_4 \sin(2t) + \frac{1}{5} e^t$$

$$\frac{d^2 x}{dt^2} = 4y + e^t$$

$$4y = \frac{d^2 x}{dt^2} - e^t$$

$$x(t) = c_1 e^{-2t} + c_2 e^{2t} + c_3 \cos(2t) + c_4 \sin(2t) + \frac{1}{5} e^t$$

$$x'(t) = -2c_1 e^{-2t} + 2c_2 e^{2t} - 2c_3 \sin(2t) + 2c_4 \cos(2t) + \frac{1}{5} e^t$$

$$x''(t) = 4c_1 e^{-2t} + 4c_2 e^{2t} - 4c_3 \cos(2t) - 4c_4 \sin(2t) + \frac{1}{5} e^t$$

$$4y = (4c_1 e^{-2t} + 4c_2 e^{2t} - 4c_3 \cos(2t) - 4c_4 \sin(2t) + \frac{1}{5} e^t) - e^t$$

$$\cancel{4}y = \cancel{4}c_1 e^{-2t} + \cancel{4}c_2 e^{2t} - \cancel{4}c_3 \cos(2t) - \cancel{4}c_4 \sin(2t) - \frac{\cancel{4}}{5} e^t$$

$$y = c_1 e^{-2t} + c_2 e^{2t} - c_3 \cos(2t) - c_4 \sin(2t) - \frac{1}{5} e^t$$